



INNER MONGOLIA BAOTOU BAYIN WIND POWER PROJECT

Document Prepared by Longyuan (Beijing) Carbon Asset Management
Technology Co., Ltd.

Project Title	Inner Mongolia Baotou Bayin Wind Power Project
Version	02
Report ID	-
Date of Issue	30-May-2022
Project ID	355
Monitoring Period	01-January-2013 to 29-August-2018 (both days included)
Prepared By	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.
Contact	Physical address: Building C, International Investment Building. 6-9 Fuchengmen North Street, Xicheng District, Beijing, China Telephone: +861063887032 Email: 20069189@ceic.com

CONTENTS

1	PROJECT DETAILS.....	3
1.1	Summary Description of the Implementation Status of the Project	3
1.2	Sectoral Scope and Project Type	3
1.3	Project Proponent	4
1.4	Other Entities Involved in the Project	4
1.5	Project Start Date	4
1.6	Project Crediting Period	4
1.7	Project Location	4
1.8	Title and Reference of Methodology	5
1.9	Participation under other GHG Programs.....	5
1.10	Other Forms of Credit.....	6
1.11	Sustainable Development Contributions	6
2	SAFEGUARDS	9
2.1	No Net Harm	9
2.2	Local Stakeholder Consultation	10
2.3	AFOLU-Specific Safeguards	14
3	IMPLEMENTATION STATUS	15
3.1	Implementation Status of the Project Activity	15
3.2	Deviations	16
3.3	Grouped Projects	16
4	DATA AND PARAMETERS.....	17
4.1	Data and Parameters Available at Validation	17
4.2	Data and Parameters Monitored.....	17
4.3	Monitoring Plan	18
5	QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	22
5.1	Baseline Emissions	22
5.2	Project Emissions	27
5.3	Leakage.....	27
5.4	Net GHG Emission Reductions and Removals.....	27

1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Inner Mongolia Baotou Bayin Wind Power Project (hereafter referred as the project) is a grid connected renewable energy project developed by Longyuan (Baotou) Wind Power Co., Ltd. The project is located in Damao United Banner (County), Baotou City, Inner Mongolia Autonomous Region, China. The purpose of the project is to generate electricity using wind power resources in the project region and to sell into North China Power Grid (NCPG) which is predominated by fossil fuel-fired power plants. The project involves the installation of 134 sets of Goldwind 77/1500kW wind turbines, for a total installed capacity of 201 MW.

The implementation of the project is in compliance with the registered PD. The Construction start permission date is 01-July-2007. All of the wind turbines had been fully operational since 13-April-2009, for a total installed capacity of 201 MW. In which, the First wind turbine started trail operation on 30-August-2008¹, and the project started full operation on 13-April-2009. All of the meters had been installed before the operation period, and the management structure and training system has been finished.

The project is expected to supply 465,920MWh renewable electricity annually to the power grid during the 20 years lifetime. The project achieves CO₂ emission reduction by replacing electricity generated by fossil fuel fired power plant connected into NCPG. The estimated annual emission reduction of the project is 500,678 tCO₂e.

This is the second monitoring period from 01-January-2013 to 29-August-2018 under VCS program for the project. The total GHG emission reductions generated in this monitoring period are 2,109,463 tCO₂e.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 energy industries (Renewable sources).

Project type: Grid-connected wind power project.

The project is not a grouped project.

¹The first wind turbine started trail operation on 30-August-2008 and was put into official operation on 31-August-2008. Thus, start date of VCS crediting period of the project is confirmed to be 30-August-2008 on which the project starting trail operation and is in line with the VCS Standard.

1.3 Project Proponent

Organization name	Longyuan (Baotou) Wind Power Co., Ltd.
Contact person	Mr. Wang Hetao
Title	General Manager
Address	Damao town, Baotou City, Inner Mongolia Autonomous Region, China
Telephone	+864712353878
Email	hetao.wang@chnenergy.com.cn

1.4 Other Entities Involved in the Project

Organization name	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.
Role in the Project	Consultant
Contact person	Mrs. YU Shanshan
Title	General Manager
Address	Building C, International Investment Building. 6-9 Fucheng North Street, Xicheng District, Beijing, China
Telephone	+861063887032
Email	20069189@ceic.com

1.5 Project Start Date

30-August-2008 (the date that the first wind turbine started trial operation)

1.6 Project Crediting Period

30-August-2008 to 29-August-2018 (10 years 0 months) (both days included).

1.7 Project Location

The project is located in Damao United Banner (County), Baotou City, Inner Mongolia Autonomous Region of China. The central geographical coordinates of the project are north latitude 42°20'12" and east longitude 109°16'32", its altitude is 1,580 m above sea level. The wind farm of the project is 260 km north from Baotou City. The detailed location of the project is shown in figure 1.

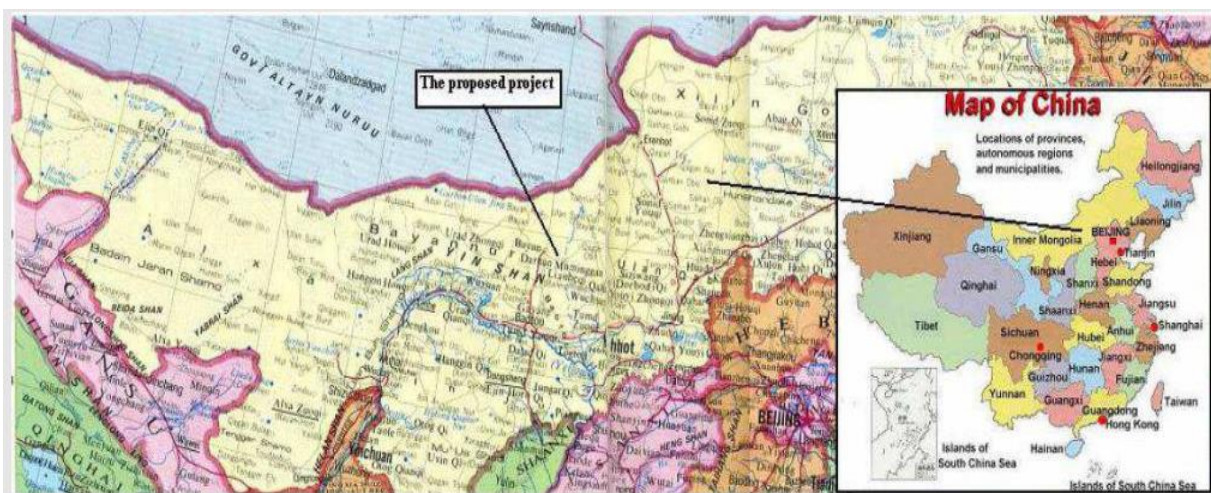


Figure 1: The detailed location of Inner Mongolia Baotou Bayin Wind Power Project

1.8 Title and Reference of Methodology

Applied methodology: ACM0002 (Version 6): Consolidated methodology for grid-connected electricity generation from renewable sources.

Applied tools: Tool for the Demonstration and Assessment of Additionality (version 04).

The applied methodology and tool can be found from the following link:

<https://cdm.unfccc.int/methodologies/PAmethodologies/approved>

1.9 Participation under other GHG Programs

The project is a registered CDM project with reference No. 2153, the detailed information can be found at <https://cdm.unfccc.int/Projects/DB/DNV-CUK1218624488.68/view>.

Start Date	End Date	GHG Emission Reduction (tCO ₂ e)	Credit	Program
30-August-2008	11-May-2009	173,742	VCU	VCS
12-May-2009	30-June-2009	86,884	CER	CDM
01-July-2009	31-December-2009	247,760	CER	CDM
01-January-2010	30-April-2010	166,957	CER	CDM
01-May-2010	30-November-2010	303,213	CER	CDM

01-December-2010	30-April-2011	208,839	CER	CDM
01-May-2011	30-April-2012	408,447	CER	CDM
01-May-2011	31-December-2012	283,359	CER	CDM

This is the second monitoring period under VCS program from 01-January-2013 to 29-August-2018 for the project. All emission reductions during this monitoring period have not and will not seek CDM CER issuance. Emission reductions during this monitoring period will only seek issuance under VCS program.

1.10 Other Forms of Credit

Emission Trading Programs and Other Binding Limits

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO₂e/year, not including renewable project². Thus, the project proponent: Longyuan (Baotou) Wind Power Co., Ltd., as an enterprise for renewable energy investment, is not included in the compliance entity list by China national Emission Trading Scheme (ETS). Furthermore, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction under the China's ETS.

Therefore, the project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. The net GHG emission reductions generated during this monitoring period have not been used for compliance under such programs or mechanisms. Furthermore, a statement on no double counting will be submitted to Verra to confirm the credits during this monitoring period has not been counted and will not be counted under emission trading programs and other binding limits.

Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period.

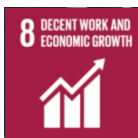
1.11 Sustainable Development Contributions

The project activity is a renewable wind power project located in Inner Mongolia Autonomous Region.

² http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html



The project utilizes wind resources to generate and supply 1,963,024.289 MWh renewable electricity to the power grid during this monitoring period, which contributes to SDG 7.



The project provides 27 long-term job opportunities for local residents including 14 female employees and the average salary rate of the project's employees is higher than the local average salary rate which contributes to SDG 8.



The renewable electricity generated by the project replaces electricity generation by fossil-fuel power plants of the power grid, which reduces 2,109,463 tCO_{2e} GHG emission during this monitoring period and contributes to SDG 13.

Since the project contributions with all impacts do not include in previously approved VCS monitoring reports or Sustainable Development Contribution Reports, all impacts during the previous monitoring period from 30/08/2008 to 31/12/2012 has not been calculated in the cumulative impact. Thus, the cumulative impacts in project lifetime are same as the one in this monitoring period.

Please refer to section 4 and section 5 of this monitoring report for the evidence of the project's contribution to SDG 7 and SDG 13. And the evidence for SDG 8 will be submitted as a separate document to Verra due to confidentiality requirement of the project proponent.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	Renewable energy supplied to the power grid	Implemented activities to increase	1,963,024.289 MWh of electricity from renewable sources has been supplied to the power grid during the reporting period	An accumulated 1,963,024.289 MWh of electricity from renewable sources has been supplied to the power grid by end of the reporting period
2)	8.5	Numbers of decent job opportunities provided by implementation of the project	Implemented activities to increase	27 people have been employed by the project as long-term employee which have an income rate higher than the local average level,	27 people have been employed by the project as long-term employee by end of the reporting period
3)	13.0	Tonnes of greenhouse gas emissions avoided	Implemented activities to increase	The project has achieved a total GHG emission reduction of 2,109,463 tCO ₂ e during the reporting period.	An accumulated 2,109,463 tCO ₂ e of GHG emission reduction have been achieved by the project by end of the reporting period

2 SAFEGUARDS

2.1 No Net Harm

The Environmental Impact Assessment Report (EIA) was approved on November 2006 by Inner Mongolia Environmental Protection Bureau.

The analysis of environmental impacts in construction phase and operation phase for the proposed project are as following:

Air

The major air pollution sources in construction phase are dust generated from constructing building foundations, tower foundations, exhaust gas from vehicles and other emissions from ordinary workers' activities. Although the construction area is dispersed, the pollution sources are intermittent, mobile and small in intensity, several measures will still be taken in order to control and reduce the air pollution. Water should be sprinkled on the mounds of soil exposed in the air to prevent the dust blowing. The soil dug from groundwork should be transported without delay. No overloading of the vehicles should be permitted to avoid any material falling down along the way. There are no air pollution sources in operation phase for the proposed project.

Water

The major wastewater emissions are mainly from domestic sewage. The domestic sewage both in construction phase and operation phase will be treated in a 50 m³ septic tank in the project site. The treated water can be discharged or recycled for watering the vegetation in project district.

Solid Waste

The main solid waste generated by the proposed project will be the construction waste and domestic waste. The construction waste will be used for backfilling tower and road foundations. The domestic wastes will, after collection, be transported to the designated waste disposal sites for centralized disposal. It is estimated that no adverse impact will be caused by the solid waste on the surrounding environment.

Noise

Noise will be generated by vehicles and machine equipment during construction. Measures will be taken to reduce the environmental impacts of noise to the maximum extent: choosing low-noise machines and vehicles, strengthening the maintenance of equipment, and enhancing construction management. Wind turbine units will produce noise of about 95-100dB(A) in operation phase, Noise attenuation calculation shows that at a distance of 250m from the sound source, noise decreases to 45dB(A), which satisfies well the Class 1 requirement of Standard of Environmental Noise of Urban Area (GB 3096-93). In addition, the residential area is far away from the wind farm, and the background noise in the proposed project site is high. So, the noise of the project will have no impact on nearby residents. For the operation members on the wind farm, some noise partition measures will be taken to reduce the noise impacts.

Birds and endangered species

There are no migratory birds / endangered species in the region of the project activity. In addition, the flight height (150m) of migratory birds is much higher than the sweeping wind height (100m) of the turbines, and, the proposed project is not located on routes of migratory birds, therefore, the proposed project activity has no negative influence on birds and endangered species.

Ecological aspects

The implementation of the proposed project will occupy some land and spoil some of the vegetation which will cause the soil erosion and have negative impact on ecological environment. A series of soil and water conservation measures will be taken to control this impact: big machinery are not allowed to enter the construction site during the construction period, earth excavation by manual labor and backpack as fully as possible that will reduce the damage to grass caused by machinery, ensuring the right placement of the machinery and other facilities, the surface soil will be preserved for backfilling during cable burying, the vehicles can only go along exclusive ways, green belt development will be undertaken all-round the project site, the temporary requisitioned land will be restored after the completion of construction by taking corresponding vegetation restoration measures, therefore, the construction of the project will put little impact on variety of local vegetation.

Socio-economic impacts

During the construction and operation of this project, the project provides employment opportunities for local residents. Therefore, the project has played an important role in the local social and economic development.

In conclusion, being as a typical type of clean renewable energy, the proposed project has no significant impacts on local environment and will greatly contribute to achievement of sustainable development objective.

2.2 Local Stakeholder Consultation

LSC prior to the project implementation

The comments on the project activity by the local stakeholders have been invited and complied in two ways:

1) Symposium

On July 10, 2007, under the support of the local government, the project participants successfully held a stakeholder meeting in Damao United Banner. Totally 28 stakeholder representatives participated the meeting, respectively from the local Environmental Protection Bureau, Local Development and Reform Commission, local Electricity Supply Company and Mingan Town. The project participants informed them about the project, asked for their comments on the project concerning socio-economic and environmental aspects, namely as follows,

- 1) Impact on the economic aspect, including the local economy, income and employment, etc;
- 2) Impact on the environmental aspect, including the ecological environment, air, noise and the impact of soil erosion, etc;

- 3) Impact on sustainable development;
- 4) Suggestions and recommendations on the proposed project;
- 5) Attitude to the implementation of the proposed project, whether or not to support.

2) Questionnaire

The project participants have also carried out a public survey on the project in the format of questionnaire. To make sure the results more representative, all these informants came from nearby areas and covered different ages and occupations.

Summary of the comments received in the symposium:

- The proposed project has been approved by the National Development and Reform Commission and Environmental Protection Bureau of Inner Mongolia Autonomous Region, which shows that the construction and operation of the proposed project will have little impacts on the local environment.
- All the stakeholder representatives believe that the proposed project will achieve CO₂ emission reduction by replacing electricity generated by fossil fuel fired power plants connected into North China Power Grid, and the proposed project will contribute to the mitigation of the GHG emissions.
- All the stakeholder representatives believe that the proposed project will have positive impacts on economy, environment and social in local region.
- The local government highly supports the proposed project, and expects the increase of local financial incoming.
- All stakeholder representatives think that the proposed project will improve the life quality of local residents, such as, providing more employment opportunities for local residents, increasing incomes of the local residents, increasing revenues from tourism with the wind farm as a beautiful landscape, accelerate the development of local service industry, and tourism.
- All stakeholder representatives think that the proposed project activity has no significant negative impacts on local environment.
- All the stakeholder representatives support and welcome the proposed project.

Public survey results:

Totally 28 questionnaires returned out of 28 with a 100% response rate. Of the 28 respondents, 12 persons are male and 16 persons are female; 9 persons are younger than 30, 14 persons are 30-40 years old and 5 persons are older than 40; 2 persons are of primary school, 10 persons are of high school, 16 persons are of college and above. Summary views on the questionnaires:

- 100% of the informants have known or heard of the proposed project.
- 100% of the informants live or work more than 1000 meters away from the proposed project.
- 100% of the informants are fully or basically satisfied with the local environment.

- 100% of the informants believe that the proposed project will have positive impacts on economy, environment and social in local region.
- 100% of the informants believe that the proposed project will help develop local economy and create new job opportunity.
- 100% of the informants believe that the implementation of the proposed project will be beneficial to local residents' income.
- 36% of the informants think that the project participants should pay attention to the problem of ecological environment, and 32% of the informants think that the project participants should pay attention to noise problems in the implementation of the proposed project. In this regard, the project participants will be in strict accordance with the environmental impact assessment report and take environmental protection measures to ease the impact on the environment.
- 100% of the informants believe that the proposed project has no significant impacts on local environment and will contribute to achievement of sustainable development objective.
- 100% of the informants are satisfied with the method and amount of compensation for the occupied land.
- 100% of the informants believe that the proposed project will contribute to the mitigation of the GHG emissions.
- 100% of the informants support and welcome the proposed project, no negative comments were received.

In conclusion, being as a typical type of clean renewable energy, the proposed project has no significant impacts on local environment and will greatly contribute to achievement of sustainable development objective, thus all the stakeholders give no negative comments and support and welcome the proposed project.

LSC during the operation period

Communications with Local stakeholders have been carried out at periodic intervals. Key implementation schedules and changes of the project have been communicated to the local authority, the neighbourhood committee, and local residents. The comments and suggestions from residents were collected by the local authority, if any.

The project owner carried out questionnaire survey for the local stakeholders to collect the relevant comments and suggestions every two years, e.g., in June 2013, July 2015 and May 2017 during this monitoring period. The local authority also conducts spot checks on the implementation of the project at periodic intervals as per relevant regulations. There have been no negative comments received for the project so far.

The following people are considered as the stakeholders of this project during the implementation stage.

- Residents of the nearby village.

- Staff of the project.
- Relevant administrative staff of the local government.

The project owner designed a questionnaire to collect the comments of relevant stakeholders and the major question issues are shown as below:

1. Are you satisfied with the current environmental quality of the project location?
2. What do you think are the main environmental problems in the current location of the project?
3. In your opinion, which environmental protection measures need to be strengthened during the operation period of this project?
4. What impact do you think this project activity may have on local employment and residents' income?
5. Impact on the local economic development.
6. Attitude on implementation of the project.

30 questionnaires have been handed in total and all of them have been received successfully from the stakeholder representatives, respectively from the local government, and surrounding villages. The representatives covered different gender, different ages, different occupations and different education levels. The detail information is shown as below:

Item		Percentage		
		June-2013	July-2015	May-2017
Amount of stakeholders surveyed	Male	50%	40%	60%
	Female	50%	60%	40%
Age	<25	6.67%	0%	3.33%
	25-45	16.67%	16.67%	16.67%
	45-55	66.66%	66.66%	16.67%
	>55	10%	16.67%	63.33%
Education	Junior high school or below	50%	40%	60%
	Senior high school	50%	50%	40%
	College or above	0%	10%	0%
Occupation	Worker	26.67%	16.67%	26.67%
	Farmer	26.67%	63.33%	26.67%
	Management personnel	6.66%	3.33%	6.66%
	Unspecified	40%	16.67%	40%

All local stakeholders are supportive to the implementation of the project, believing that the Project help improve the life of local people and promote local economic development without adverse environmental impact.

However, during the operation, some main issues were raised by local stakeholder and the developer should pay attention and solve these issues which are listed as following:

1. Some measurements should be taken to control the environmental pollution;
2. Project developer should ensure the security of the power station during the operation period;

For the first issue, as the project merely caused environmental pollution during the operation period, this issue can be easily controlled; for the second issue, during this monitoring period, all the workers received the safety training regularly.

Meanwhile, the local authority has also conducted spot checks on the implementation of the project from time to time as per the request from the local governments' regulations. There are no negative comments received for the project. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

The project is a non-AFOLU project, this section is not required.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The project activity was to install and operate 134 wind turbines, each of which has a capacity of 1500 kW. The total installed capacity is 201MW. The wind turbines are made by Xinjiang Gold wind Science and Technology Co. Ltd. This kind of wind turbine is a pitch regulated and active yaw wind turbine. The generator is a directly driven permanent magnet synchronous generator. According to a Suitability Study Report by the manufacturer, Gold wind 77/1500 kW is entirely suitable to the region. As stipulated in the Equipment Purchase Agreement (EPA), the manufacturer provides the technical training and maintenance for the project. The main technical specifications of Gold wind 77/1500kW are as follows:

Parameter	Unit	Amount
Rated Capacity	kW	1500
Rotor diameter	m	77
Blade number	-	3
Area swept	m ²	3850
Cut-in wind speed	m/s	3
Rated wind speed	m/s	11.8
Cut-out wind speed	m/s	22
Survival speed	m/s	59.5
Height of hub	m	65
Rotor speed	rpm	9-20
Temperature	°C	-30~+40
Regulating method	-	Pitch regulated and active yaw wind turbine
Generator type	-	Directly driven permanent magnet synchronous generator
Generator voltage	V	690

A new 35/220kV transformer station is constructed within the project. Each turbine has a 690V to 35kV transformer, from which a 35kV line linked into on-site 220kV transformer at the station, through the 220kV transformer the electricity output is transmitted to Wanghai 220kV substation, and then to Inner Mongolia Western Power Grid which is an integral part of North China Power Grid. The auxiliary electric system of the project includes on-site control, protection, measure, signaling and surveillance in central control room at the site. The targets to be controlled and monitored include 134 wind turbines and transformers. The wind farm is dispatched by regional dispatch centre and wind turbines could be measured and signaled remotely by Internet.

During this monitoring period, the project was implemented in accordance with the registered VCS PD. The Project has operated without any accidental or emergency events that might impact the GHG emission reductions or removals and monitoring.

3.2 Deviations

3.2.1 Methodology Deviations

There is no methodology deviation in this monitoring period.

3.2.2 Project Description Deviations

The project is registered under VCS Version 2007.1 and completed validation before 19/03/2020. Thus, it remains eligible to apply the crediting period requirements under VCS version 2007.1 which shall be a maximum of ten years and may be renewed at most twice, the 1st crediting period of the project should be updated from “30-August-2008 to 11-May-2009” to “30-August-2008 to 29-August-2018”.

The project was registered under CDM (Reference number: 2153) on 12-May-2009 (7 years x 3, renewable). According to the registered CDM PDD, the measurement for the electricity imported via 10kV back up line has been integrated into the monitoring activity of the Project. However, based on the clarification of the actual situation from PP, the 10kV back up line has been deactivated on 25-December-2012 and no more electricity was imported through this 10kV back up line. Thus, the net electricity supplied to the grid (EG_y) is the same as the net electricity supplied to the grid through the main line (EG'_y), i.e. $EG_y = EG'_y$, which is calculated as the difference between the electricity exports and imports through main line. Therefore, the parameter $EG_{\text{backupline}}$ and EG'_y have been removed from section 4.2. The detail information of the data and parameters monitored, refer to the Section 4.2.

The deviation for the crediting period and monitoring plan has no impacts on the applicability of the methodology, additionality or baseline of the project, and the accuracy and conservativeness of monitoring plan and ER calculation. There are no other project description deviations occurred in this monitoring period or a previous monitoring period.

3.3 Grouped Projects

Not applicable as this is not a grouped project

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _y
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor of the power grid in year y
Source of data	Registered CDM PDD
Value applied	1.07460
Justification of choice of data or description of measurement methods and procedures applied	Determined ex-ante and fixed for the 1 st crediting period
Purpose of Data	Calculation of baseline emissions
Comments	Determined ex-ante and fixed for the 1 st crediting period

4.2 Data and Parameters Monitored

Data / Parameter	EG _y				
Data unit	MWh				
Description	Net electricity supplied by the project activity to the grid during the year y				
Source of data	Reading records from Meter1/Meter2				
Description of measurement methods and procedures to be applied	Continuously measurement, daily recorded and monthly aggregated using the calibrated meter				
Frequency of monitoring/recording	Continuously measured, daily recorded and monthly aggregated.				
Value monitored	1,963,024.289				
Monitoring equipment	Monitoring meter	Meter 1	Meter 1' (Backup meter of Meter1)	Meter 2	Meter 2' (Backup meter of Meter2)
	Type	ZMQ202C.4r4af6	ZMQ202C.4r4af6	ZMQ202C.4r4af6	ZMQ202C.4r4af6
	Accuracy	0.2S			
	Serial No.	94471210	94471211	95151114	95151118
	Calibration frequency	Annually			

	Calibration date	10-November-2012 08-November-2013 06-November-2014 02-November-2015 30-October-2016 26-October-2017
	Calibration validity	09-November-2013 07- November -2014 05- November -2015 01- November -2016 29-October-2017 25-October-2018
QA/QC procedures to be applied	The bi-directional meters with accuracy degree of 0.2S used for measuring are calibrated annually as per industry standards of host country (DL/T448). Meter Readings were double checked by ETNs or electricity confirmation letter by grid company. All data and records to be achieved for at least 2 years after the end of last crediting period.	
Purpose of the data	Calculation of baseline emissions	
Calculation method	The EG_y is calculated as the electricity exported to the grid via the main line minus the electricity imported from the grid via the main line.	
Comments	-	

4.3 Monitoring Plan

Monitoring tasks must be implemented according to the monitoring plan in order to ensure that the real, measurable and long-term greenhouse gas (GHG) emission reduction for the proposed project is monitored and reported.

This monitoring plan is designed for Inner Mongolia Baotou Bayin Wind Power Project implemented by Longyuan (Baotou) Wind Power Co., Ltd (the project owner).

1. Organizational structure

The organizational structure and responsibilities of the personnel is shown as follows:

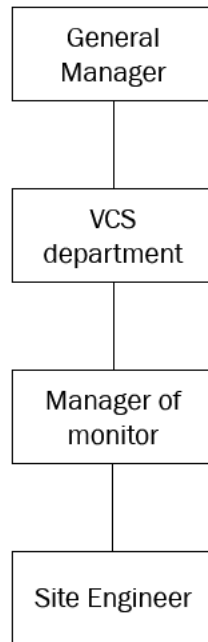


Figure 1 Organizational structure

The general manager is in charge of decision-making on major VCS issues, the VCS department is responsible for the concrete implementation of the Monitoring Plan. The monitor manager is responsible for supervising and checking the whole data record process and the calibration of meters. Another main task of the monitor manager is facilitating the verification through providing the VVB with all required necessary information. The site engineers will collect monitoring data (e.g. electric meter data) and keep the ETNs, etc. Before the start of the monitor, all members in the VCS department should have received well training on VCS issues.

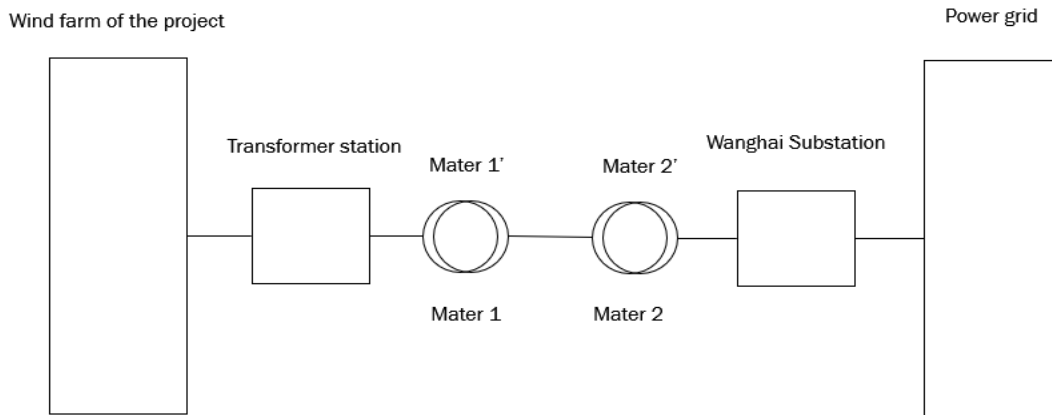
2. Monitoring equipment

The first metering equipment (Meter1 and its backup Meter1') is installed at the outlet of the transformer station (35kV/220kV) of the project.

The second metering equipment (Meter2 and its backup Meter2') is installed at the entrance to Wanghai 220 kV substation of the grid.

The meters are the multifunctional electricity meters (accuracy degree is 0.2S, bidirectional) capable of recording import and export of electricity and providing output in the form of electricity supplied to the grid.

The installation of electricity meters is shown in the figure below:



3. Data recording

The project owner is responsible for the operation of the first metering equipments (Meter1 and Meter1'), and Grid Company is responsible for the operation of the second metering equipment (Meter2 and Meter2') to ensure that all meters are in good conditions and the net electricity supplied to the grid is valid and is able to achieve emission reductions calculation. Under such circumstances, the specific steps for data collection and reporting are listed below:

- 1) Power Grid Company read the second metering equipments (Meter2 and Meter 2'), and the project owner read the first metering equipments (Meter 1 and Meter 1') respectively and record readings at 00:00 daily and aggregated monthly. According to the requirement of local grid company, the cut-off time of electricity exports and imports is at 0:00 on 21st day of each month.
- 2) Power Grid Company supplies readings to the project owner and provides relevant documents.
- 3) Project owner checks the readings from the grid company with readings of Meter1 or Meter1', and after confirmed by both sides, the grid company issues the ETNs on confirmed readings of electricity exports and imports based on the readings of Meter2 or Meter2'.
- 4) Project owner provides the meter's readings and copies of ETNs to VVB for verification.

Should any previous months reading of the meter be inaccurate by more than the allowable error, or otherwise functioned improperly, the electricity generation supplied to the grid by the proposed project shall be determined by:

- 1) The project owner and Power Grid Company shall jointly prepare an estimate of the correct reading; and, provide ample evidence to VVB that the method is reasonable;
- 2) The backup meter reading shall be for backup usage;
- 3) If the project owner and Power Grid Company fail to agree the estimate of the correct reading, then the matter will be referred for arbitration according to agreed procedures.

During this monitoring period, all meters are working well, thus the meter readings applied for ER calculations are sourced from Meter1 and Meter2 and ETNs issued by grid company were based on the reading records of Meter2.

4. Quality Assurance and Quality Control

The project activities will use high-precision monitoring equipment to monitor the electricity from and to the grid. Necessary backup meters will be installed, to operate for backup usage when the relevant meter is out of order. All meters will be calibrated and sealed as per the industry practices at regular intervals. Hence, high quality is ensured. ETNs are used to test the consistency of the recorded data.

All parameters monitored under the monitoring plan will be archived electronically and be kept at least for 2 years after the end of last crediting period. The monitored data will be presented to the verification agency or VVB to whom verification of emission reductions is assigned.

The documents in paper format, such as maps, tables, and the EIA report, will be used in conjunction with the monitoring plan to check the authenticity of the information, and be kept at least one copy by the project owner

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

As per the registered PD, the baseline emission of the project is calculated as below:

$$BE_y = EG_y \times EF_y \quad (1)$$

Where,

BE_y Baseline emissions in year y (t CO₂e)

EG_y Net electricity delivered to the grid by the project (MWh)

EF_y Baseline emission factor in the year y (tCO₂e/MWh)

The monitored EG_y is reported in the table below.

Table 1 Monitored monthly value of EG_y

Year	Month	Reading records of electricity exported to the grid via the main line from Meter1 (MWh)	Reading records of electricity exported to the grid via the main line from Meter2 (MWh)	Values of electricity exported to the grid via the main line from ETNs (MWh)	Conservative values of electricity exported to the grid via the main line (MWh)	Reading records of electricity imported from the grid via main line from Meter1 (MWh)	Reading records of electricity imported from the grid via main line from Meter2 (MWh)	Values of electricity imported from the grid via the main line from ETNs (MWh)	Conservative values of electricity imported from the grid via the main line (MWh)	EG _y (MWh)
		A	B	C	D=min (A,B,C)	E	F	G	H=max(E,F,G)	I=D-H
2013	01-Jan~20-Jan	22,923.770	22,915.110	22,915.110	22,915.110	103.110	105.120	105.120	105.120	22,809.990
	21-Jan~20-Feb	29,442.642	29,424.430	29,424.430	29,424.430	116.510	117.740	117.740	117.740	29,306.690
	21-Feb~20-Mar	37,280.622	37,262.410	37,262.410	37,262.410	106.510	107.740	107.740	107.740	37,154.670
	21-Mar~20-Apr	57,146.872	57,128.660	57,128.660	57,128.660	144.460	145.690	145.690	145.690	56,982.970
	21-Apr~20-May	39,404.895	39,383.895	39,383.895	39,383.895	95.280	96.510	96.510	96.510	39,287.385
	21-May~20-Jun	45,020.882	45,002.670	45,002.670	45,002.670	101.420	102.650	102.650	102.650	44,900.020
	21-Jun~20-Jul	27,537.272	27,519.060	27,519.060	27,519.060	184.760	185.990	185.990	185.990	27,333.070
	21-Jul~20-Aug	37,834.912	37,816.700	37,816.700	37,816.700	102.220	103.450	103.450	103.450	37,713.250
	21-Aug~20-Sept	39,402.795	39,383.895	39,383.895	39,383.895	168.590	169.820	169.820	169.820	39,214.075
	21-Sept~20-Oct	42,357.522	42,339.310	42,339.310	42,339.310	77.060	78.290	78.290	78.290	42,261.020
	21-Oct~20-Nov	29,227.472	29,209.260	29,209.260	29,209.260	123.590	124.820	124.820	124.820	29,084.440
	21-Nov~20-Dec	21,403.232	21,385.020	21,385.020	21,385.020	209.660	210.890	210.890	210.890	21,174.130
	21-Dec~31-Dec	12,820.100	12,814.546	12,814.546	12,814.546	44.461	44.930	44.930	44.930	12,769.616
	Subtotal	441,802.988	441,584.966	441,584.966	441,584.966	1,577.631	1,593.640	1,593.640	1,593.640	439,991.326
2014	01-Jan~20-Jan	23,309.273	23,299.174	23,299.174	23,299.174	80.838	81.690	81.690	81.690	23,217.484
	21-Jan~20-Feb	25,108.273	25,092.620	25,092.620	25,092.620	238.609	239.930	239.930	239.930	24,852.690
	21-Feb~20-Mar	34,941.123	34,925.470	34,925.470	34,925.470	90.889	92.210	92.210	92.210	34,833.260
	21-Mar~20-Apr	27,348.373	27,332.720	27,332.720	27,332.720	150.179	151.500	151.500	151.500	27,181.220
	21-Apr~20-May	46,056.643	46,040.990	46,040.990	46,040.990	136.019	137.340	137.340	137.340	45,903.650

	21-May~20-Jun	41,338.383	41,322.730	41,322.730	41,322.730	137.939	139.260	139.260	139.260	41,183.470
	21-Jun~20-Jul	24,079.703	24,064.050	24,064.050	24,064.050	195.699	197.020	197.020	197.020	23,867.030
	21-Jul~20-Aug	24,296.023	24,280.370	24,280.370	24,280.370	24.247	25.568	25.568	25.568	24,254.802
	21-Aug~20-Sept	19,020.870	19,006.890	19,006.890	19,006.890	230.759	232.080	232.080	232.080	18,774.810
	21/-Sept~20-Oct	36,410.883	36,395.230	36,395.230	36,395.230	137.779	139.100	139.100	139.100	36,256.130
	21-Oct~20-Nov	39,572.323	39,556.670	39,556.670	39,556.670	120.969	122.290	122.290	122.290	39,434.380
	21-Nov~20-Dec	54,077.303	54,061.650	54,061.650	54,061.650	41.689	43.010	43.010	43.010	54,018.640
	21-Dec~31-Dec	14,461.733	14,455.629	14,455.629	14,455.629	23.745	24.214	24.214	24.214	14,431.415
	Subtotal	410,020.906	409,834.193	409,834.193	409,834.193	1,609.360	1,625.213	1,625.213	1,625.213	408,208.980
2015	01-Jan~20-Jan	26,294.059	26,282.961	26,282.961	26,282.961	43.172	44.026	44.026	44.026	26,238.935
	21-Jan~20-Feb	23,347.362	23,330.160	23,330.160	23,330.160	128.787	130.110	130.110	130.110	23,200.050
	21-Feb~20-Mar	29,334.242	29,317.040	29,317.040	29,317.040	90.567	91.890	91.890	91.890	29,225.150
	21-Mar~20-Apr	36,445.662	36,428.460	36,428.460	36,428.460	103.347	104.670	104.670	104.670	36,323.790
	21-Apr~20-May	50,926.312	50,909.110	50,909.110	50,909.110	135.837	137.160	137.160	137.160	50,771.950
	21-May~20-Jun	44,347.872	44,330.670	44,330.670	44,330.670	93.227	94.550	94.550	94.550	44,236.120
	21-Jun~20-Jul	21,781.692	21,764.490	21,764.490	21,764.490	201.317	202.640	202.640	202.640	21,561.850
	21-Jul~20-Aug	25,669.432	25,652.230	25,652.230	25,652.230	181.047	182.370	182.370	182.370	25,469.860
	21-Aug~20-Sept	24,595.672	24,578.470	24,578.470	24,578.470	205.267	206.590	206.590	206.590	24,371.880
	21/-Sept~20-Oct	28,268.402	28,251.200	28,251.200	28,251.200	105.877	107.200	107.200	107.200	28,144.000
	21-Oct~20-Nov	32,129.982	32,112.780	32,112.780	32,112.780	188.397	189.720	189.720	189.720	31,923.060
	21-Nov~20-Dec	27,519.202	27,502.000	27,502.000	27,502.000	195.037	196.360	196.360	196.360	27,305.640
	21-Dec~31-Dec	8,920.294	8,915.599	8,915.599	8,915.599	49.378	49.894	49.894	49.894	8,865.705
	Subtotal	379,580.185	379,375.171	379,375.171	379,375.171	1,721.257	1,737.180	1,737.180	1,737.180	377,637.991
2016	01-Jan~20-Jan	16,218.716	16,210.181	16,210.181	16,210.181	89.778	90.716	90.716	90.716	16,119.465
	21-Jan~20-Feb	26,613.060	26,599.830	26,599.830	26,599.830	60.796	62.250	62.250	62.250	26,537.580
	21-Feb~20-Mar	25,531.520	25,518.290	25,518.290	25,518.290	46.666	48.120	48.120	48.120	25,470.170
	21-Mar~20-Apr	28,479.910	28,466.680	28,466.680	28,466.680	73.666	75.120	75.120	75.120	28,391.560
	21-Apr~20-May	21,821.914	21,809.560	21,809.560	21,809.560	89.386	90.840	90.840	90.840	21,718.720

	21-May~20-Jun	22,342.360	22,329.130	22,329.130	22,329.130	143.326	144.780	144.780	144.780	22,184.350
	21-Jun~20-Jul	20,878.600	20,865.370	20,865.370	20,865.370	105.836	107.290	107.290	107.290	20,758.080
	21-Jul~20-Aug	14,869.920	14,856.690	14,856.690	14,856.690	144.166	145.620	145.620	145.620	14,711.070
	21-Aug~20-Sept	13,052.190	13,038.960	13,038.960	13,038.960	180.506	181.960	181.960	181.960	12,857.000
	21/-Sept~20-Oct	16,525.900	16,512.670	16,512.670	16,512.670	97.966	99.420	99.420	99.420	16,413.250
	21-Oct~20-Nov	22,179.880	22,166.650	22,166.650	22,166.650	69.406	70.860	70.860	70.860	22,095.790
	21-Nov~20-Dec	23,288.590	23,275.360	23,275.360	23,275.360	122.376	123.830	123.830	123.830	23,151.530
	21-Dec~31-Dec	5,315.825	5,310.782	5,310.782	5,310.782	37.739	38.216	38.216	38.216	5,272.566
	Subtotal	257,118.385	256,960.153	256,960.153	256,960.153	1,261.613	1,279.022	1,279.022	1,279.022	255,681.131
2017	01-Jan~20-Jan	9,665.135	9,655.968	9,655.968	9,655.968	68.616	69.484	69.484	69.484	9,586.484
	21-Jan~20-Feb	20,822.080	20,807.870	20,807.870	20,807.870	75.165	76.510	76.510	76.510	20,731.360
	21-Feb~20-Mar	17,447.300	17,433.090	17,433.090	17,433.090	103.495	104.840	104.840	104.840	17,328.250
	21-Mar~20-Apr	23,384.030	23,369.820	23,369.820	23,369.820	99.505	100.850	100.850	100.850	23,268.970
	21-Apr~20-May	22,471.200	22,456.990	22,456.990	22,456.990	120.755	122.100	122.100	122.100	22,334.890
	21-May~20-Jun	19,204.810	19,190.600	19,190.600	19,190.600	84.495	85.840	85.840	85.840	19,104.760
	21-Jun~20-Jul	18,553.480	18,539.270	18,539.270	18,539.270	192.555	193.900	193.900	193.900	18,345.370
	21-Jul~20-Aug	16,755.110	16,740.900	16,740.900	16,740.900	200.455	201.800	201.800	201.800	16,539.100
	21-Aug~20-Sept	21,551.070	21,536.860	21,536.860	21,536.860	165.225	166.570	166.570	166.570	21,370.290
	21/-Sept~20-Oct	26,631.270	26,617.060	26,617.060	26,617.060	123.705	125.050	125.050	125.050	26,492.010
	21-Oct~20-Nov	42,839.840	42,825.630	42,825.630	42,825.630	89.045	90.390	90.390	90.390	42,735.240
	21-Nov~20-Dec	49,999.810	49,985.600	49,985.600	49,985.600	107.665	109.010	109.010	109.010	49,876.590
	21-Dec~31-Dec	9,613.209	9,605.399	9,605.399	9,605.399	41.171	41.569	41.569	41.569	9,563.829
	Subtotal	298,938.344	298,765.056	298,765.056	298,765.056	1,471.852	1,487.913	1,487.913	1,487.913	297,277.143
2018	01-Jan~20-Jan	17,478.561	17,464.361	17,464.361	17,464.361	74.856	75.581	75.581	75.581	17,388.781
	21-Jan~20-Feb	29,167.190	29,145.180	29,145.180	29,145.180	114.457	115.580	115.580	115.580	29,029.600
	21-Feb~20-Mar	21,190.440	21,168.430	21,168.430	21,168.430	95.607	96.730	96.730	96.730	21,071.700
	21-Mar~20-Apr	26,070.400	26,048.390	26,048.390	26,048.390	80.567	81.690	81.690	81.690	25,966.700
	21-Apr~20-May	24,396.580	24,376.580	24,376.580	24,376.580	52.467	53.590	53.590	53.590	24,322.990

21-May~20-Jun	24,203.000	24,180.990	24,180.990	24,180.990	138.647	139.770	139.770	139.770	24,041.220
21-Jun~20-Jul	21,275.670	21,253.660	21,253.660	21,253.660	325.207	326.330	326.330	326.330	20,927.330
21-Jul~20-Aug	15,360.540	15,338.540	14,348.956	14,348.956	89.796	90.892	90.892	90.892	14,258.064
21-Aug~29-Aug	7,255.341	7,249.541	7,249.541	7,249.541	27.372	28.208	28.208	28.208	7,221.333
Subtotal	186,397.723	186,225.672	185,236.088	185,236.088	998.976	1,008.370	1,008.370	1,008.370	184,227.718
Total	1,973,858.530	1,972,745.211	1,971,755.627	1,971,755.627	8,640.689	8,731.338	8,731.338	8,731.338	1,963,024.289

***Note:**

1. The most conservative values among readings from Meter1, Meter2 and values from ETNs are applied for ER calculations.
2. Readings for the period 01-Jan to 20-Jan in 2013 to 2018 and 21-Dec to 31-Dec in 2013 to 2017 as well as readings for 21-Aug to 29-Aug in 2018 were sourced from the daily reading records and corresponding confirmation has been issued by grid company.

The baseline emissions of the project are calculated as follows:

Table 2 Calculation of Baseline emissions

Year	EG_y (MWh)	EF_y (tCO ₂ /MWh)	BE_y (tCO ₂)
01-January-2013 to 31-December-2013	439,991.326	1.07460	472,814
01-January-2014 to 31-December-2014	408,208.980	1.07460	438,661
01-January-2015 to 31-December-2015	377,637.991	1.07460	405,809
01-January-2016 to 31-December-2016	255,681.131	1.07460	274,754
01-January-2017 to 31-December-2017	297,277.143	1.07460	319,454
01-January-2018 to 29-August-2018	184,227.718	1.07460	197,971
Total	1,963,024.289	-	2,109,463

5.2 Project Emissions

As per the applied methodology, the project emission is zero.

5.3 Leakage

As per the applied methodology, leakage of the project is not considered.

5.4 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01-January-2013 to 31-December-2013	472,814	0	0	472,814
01-January-2014 to 31-December-2014	438,661	0	0	438,661
01-January-2015 to 31-December-2015	405,809	0	0	405,809
01-January-2016 to 31-December-2016	274,754	0	0	274,754

01-January-2017 to 31-December- 2017	319,454	0	0	319,454
01-January-2018 to 29-August-2018	197,971	0	0	197,971
Total	2,109,463	0	0	2,109,463

Comparison of actual emission reduction and the estimation in the registered PD.

Monitoring period	Emission reduction achieved (tCO ₂)	Estimated ER in registered PD (tCO ₂)	Difference
01-January-2013 to 29- August-2018	2,109,463	2,835,346	-25.60%

The actual emission reduction achieved during this monitoring period is 25.60% lower than the estimation for the same period in the registered CDM-PDD due to the fluctuation in wind resources and limitation of the grid absorption capacity. Refer to ER calculation spreadsheet for detailed calculation process.

APPENDIX: <SUPPORTING EVIDENCE>

Evidence for SDG 7 and SDG 13: From project operation date onwards, the net electricity supplied to the grid and corresponding emission reduction in every monitoring period have been listed in below table. Please refer to following links for details.

<https://cdm.unfccc.int/Projects/DB/DNV-CUK1218624488.68/view>

<https://registry.verra.org/app/projectDetail/VCS/355>

Start Date	End Date	EG _y (MWh)	ER (tCO ₂ e)
30-August-2008	11-May-2009	161,680.625	173,742
12-May-2009	30-June-2009	81,102.53	86,884
01-July-2009	31-December-2009	217,173.157	247,760
01-January-2010	30-April-2010	155,367.396	166,957
01-May-2010	30-November-2010	282,164.268	303,213
01-December-2010	30-April-2011	194,341.948	208,839
01-May-2011	30-April-2012	380,189.52	408,447
01-May-2011	31-December-2012	263,688.10	283,359
01-January-2013	29-August-2018	1,963,024.289	2,109,463

Evidence for SDG 8: The employee list of the project has been submitted as a separate document to VVB due to confidentiality requirement of the project proponent.